

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I N. Tharun Sai (192472382) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	

		and requirements accurately				
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s=c \log_{10}(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0 2

1 2

2 4

3 8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$[0 \ -1 \ \ 0], \{-1 \ \ \ 4 \ -1\}, \{0 \ -1 \ \ 0\}$

to the matrix:

$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$

Compute the output at the center.

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Introduction

Grey level transformation is a fundamental image enhancement technique in digital image processing. It modifies the intensity (brightness) values of pixels to improve the visual appearance of an image. By changing the grey levels, hidden details become more visible, making the image easier to interpret for both humans and computer vision systems.

Explanation

The main purposes of grey level transformation are:

- To improve image brightness and contrast.
- To highlight important image features.
- To reveal hidden details in dark or bright regions.
- To prepare images for further processing such as segmentation and edge detection.

Common grey level transformations include:

- Negative transformation
- Log transformation
- Power-law (Gamma) transformation
- Contrast stretching

(b) Apply the negative transformation $s = 255 - r$

Given:

$$r = [50, 100, 150, 200]$$

Formula:

$$s = 255 - r$$

Calculation

For **r = 50**

$$s = 255 - 50 = 205$$

For **r = 100**

$$s = 255 - 100 = 155$$

For **r = 150**

$$s = 255 - 150 = 105$$

For **r = 200**

$$s = 255 - 200 = 55$$

Answer

Original Pixel (r)	Transformed Pixel (s)
50	205
100	155

Original Pixel (r)	Transformed Pixel (s)
150	105
200	55

Conclusion

Negative transformation reverses the intensity values, making bright regions dark and dark regions bright. It is useful for enhancing white details on dark backgrounds.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Introduction

Log transformation is a grey level transformation used to expand the values of dark pixels while compressing brighter pixels. It improves the visibility of details present in low-intensity regions of an image.

Explanation

The formula is:

$$s = c \log(1 + r)$$

where:

- **r** = input pixel value
- **s** = output pixel value
- **c** = constant

It increases low-intensity values significantly while compressing high-intensity values, making dark regions brighter.

(b) Compute $s = c\log(1 + r)$, where $c = 10$

Using **log base 10**

For $r = 0$

$$s = 10\log(1) = 0$$

For $r = 10$

$$\begin{aligned}s &= 10\log(11) \\ &= 10(1.0414) \\ &= 10.41\end{aligned}$$

For $r = 50$

$$\begin{aligned}s &= 10\log(51) \\ &= 10(1.7076) \\ &= 17.08\end{aligned}$$

For $r = 100$

$$\begin{aligned}s &= 10\log(101) \\ &= 10(2.0043) \\ &= 20.04\end{aligned}$$

Answer

r	s
0	0
10	10.41

r	s
50	17.08
100	20.04

Conclusion

Log transformation brightens dark regions and improves visibility while preventing very bright regions from becoming overexposed.

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Introduction

Histogram Equalization is an image enhancement technique used to improve image contrast by redistributing the intensity values across the available range.

Explanation

It works by:

- Computing the histogram.
- Finding the cumulative distribution function (CDF).
- Mapping old intensity values to new values.
- Producing a nearly uniform histogram.

This improves contrast and reveals hidden details.

(b) Compute cumulative distribution and equalized intensity

Given Histogram

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Step 1: Cumulative Frequency

Intensity	Frequency	CDF
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalized Value

Formula

$$s = \frac{CDF}{N} (L - 1)$$

where

$$L = 4$$

Therefore

Intensity 0

$$\frac{2}{16} \times 3 = 0.375 \approx 0$$

Intensity 1

$$\frac{4}{16} \times 3 = 0.75 \approx 1$$

Intensity 2

$$\frac{8}{16} \times 3 = 1.5 \approx 2$$

Intensity 3

$$\frac{16}{16} \times 3 = 3$$

Final Answer

Intensity	CDF	Equalized Intensity
0	2	0
1	4	1

Intensity	CDF	Equalized Intensity
2	8	2
3	16	3

Conclusion

Histogram Equalization spreads intensity values more uniformly, improving image contrast and making details easier to observe.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Introduction

Spatial smoothing filters reduce noise by replacing each pixel with the average value of its neighboring pixels.

Explanation

Advantages:

- Removes random noise.
- Produces smoother images.
- Improves image quality before segmentation.

Common smoothing filters:

- Mean filter
- Gaussian filter
- Median filter

(b) Apply a 3×3 averaging filter

Matrix

$$\begin{matrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{matrix}$$

Average Filter

$$\frac{1}{9}$$

Sum

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

Answer

New center pixel

$$\boxed{30}$$

Conclusion

The averaging filter smooths the image by replacing the center pixel with the average of its neighboring pixels, reducing random noise.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Introduction

Sharpening is an image enhancement technique used to increase the visibility of edges and fine details. It emphasizes high-frequency components and improves image clarity after smoothing.

Explanation

Sharpening:

- Highlights object boundaries.
- Improves edge visibility.
- Restores image details.
- Enhances high-frequency information.

Common sharpening methods:

- Laplacian Filter
- High-pass Filter
- Unsharp Masking

(b) Apply the Laplacian Mask

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image Region

10	20	30
20	30	40
30	40	50

Calculation

$$\begin{aligned} &= (0 \times 10) + (-1 \times 20) + (0 \times 30) \\ &+ (-1 \times 20) + (4 \times 30) + (-1 \times 40) \\ &+ (0 \times 30) + (-1 \times 40) + (0 \times 50) \\ &= 0 - 20 + 0 - 20 + 120 - 40 + 0 - 40 + 0 \\ &= 120 - 120 \\ &= \boxed{0} \end{aligned}$$

Answer

The output of the Laplacian filter for the given region is:

$$\boxed{0}$$

Conclusion

The Laplacian filter detects rapid intensity changes and enhances edges by emphasizing high-frequency components. It is widely used to sharpen blurred images and improve fine details.